EEL 5764: Computer Architecture

Semester Research Project Guidelines & Evaluation Standards (Fall 2026)

Course Weight: 40% of Final Grade	**Instructor:** Dr. Sandip Ray (MALA 4131)
Release Date: Thursday, September 17, 2026	**In-Class Workshop:** Tuesday, September 22, 2026

1. Semester Project Overview & Evaluation Matrix (40% Total)

The semester research project is a central component of EEL 5764, accounting for 40% of the course grade. The project offers graduate students hands-on experience in investigating modern computer architecture problems, evaluating microarchitectural mechanisms, and communicating technical findings through professional research formats. Evaluation is broken down into four distinct milestones to ensure continuous progress and feedback:

Milestone Component	Weight	Due Date	Key Requirements
Project Group Registration	—	Tuesday, Sept 22	Finalize team members on Canvas (at most 4 students per group).
Project Proposal	15%	Thursday, Oct 1 (11:59 PM)	2-page PDF document detailing problem definition, mechanism, and methodology.
Recorded Video Presentation	30%	Tuesday, Nov 17 (11:59 PM)	5-minute recorded video presentation summarizing architecture & findings.
Canvas Peer Q&A	20%	Nov 19 – Nov 20	Asynchronous peer discussion board Q&A on submitted videos.
Final Written Report	35%	Tuesday, Dec 1 (11:59 PM)	Complete IEEE double-column conference-style research paper.

2. Group Formation & Equal Evaluation Policy

• **Group Size Constraints:** Projects must be conducted in groups of **at most 4 students** (teams of 1, 2, 3, or 4 members).

• **Equal Evaluation Policy:** All project groups will be evaluated under the exact same academic standards. There is **no extra concession or penalty** for deciding to work in a group of less than 4 (e.g., an individual or 2-person team is held to the exact same quality and depth standards as a 4-person team).

• **Canvas Group Registration:** All project groups must be officially registered and finalized in Canvas by **Tuesday, September 22**.

• **Single Submission Rule:** Exactly **one submission per group** is required for every project material (Proposal, Video, and Final Report). All group members will receive the same grade on submitted group materials.

3. Project Scope, Freedom of Topic & Preferred Simulation Platforms

• **Calibrated Project Complexity:** The active execution window for the project spans approximately **one month** (from proposal approval on October 1 to video submission in mid-November). Projects should be calibrated to a level of complexity that can be thoroughly completed within this timeframe. **The scope should not be overly conservative (e.g., simple textbook lookups), but also not overly ambitious.** Open-ended, unconstrained research is obviously not possible in a single month; focus on exploring a well-defined, interesting architectural direction.

• **Freedom of Topic Choice:** Students are completely free to choose their own project topics. All research tracks and topic ideas discussed during class lectures are strictly **suggestions—you are not bound by them!** You are encouraged to propose topics aligned with your specific research interests in computer architecture.

• **Preferred Simulation Platform (gem5):** The **gem5 simulator** is the preferred platform for exploring various microarchitectural directions. However, gem5 is **not a mandatory requirement**. Teams may utilize other simulation tools (e.g., GPU-Sim, McPAT), SystemC/HDL hardware prototypes, or analytical modeling frameworks if properly justified in their proposal.

4. Project Proposal Requirements (Due Thursday, Oct 1)

The project proposal is a 2-page formal document (PDF upload on Canvas by 11:59 PM on October 1). The proposal must clearly outline the problem you are exploring and how you plan to conduct your study. Every proposal must explicitly address the following four core components:

1. Problem Definition: What specific architectural bottleneck, memory wall constraint, power efficiency limit, or hardware security vulnerability are you exploring?

2. Proposed Mechanism / Approach: What specific architectural mechanism, policy modification, hardware structure, or design trade-off do you propose or plan to evaluate?

3. Experimental Methodology: How do you expect to explore and evaluate your mechanism? Specify your target platform (gem5 or alternative), baseline processor configuration, parameter sweeps, and benchmark workloads.

4. Target Evaluation Metrics: What quantitative metrics will you collect to measure success (e.g., CPI, IPC, AMAT, cache miss rates, energy-delay product, latency penalties)?

5. Instructor Consultation & Pre-Proposal Discussion

Students are **strongly encouraged to talk to Dr. Sandip Ray** if they have questions or want to discuss their proposal before formal submission! Seeking early feedback helps ensure your proposed scope is well-calibrated—neither too conservative nor overly ambitious. You can consult the instructor during office hours (MALA 4131), via Canvas messaging, or during the **In-Class Project Topic Workshop on Tuesday, September 22.**

6. Suggested Research Tracks (Non-Binding Examples)

To help inspire topic selection, the following research tracks outline representative project ideas. Remember that these are strictly non-binding suggestions; you are free to explore any topic in computer architecture.

Research Track Area	Representative Topic Examples
Core & Pipeline Optimization	Branch predictor comparative analysis (TAGE vs. Tournament), ROB window sizing under memory stalls, hardware forwarding path removal trade-offs.
Memory Hierarchy & Caches	Hardware prefetching algorithms (stride vs. stream), L2 cache replacement policies (LRU/LFU/Random), MSHR depth & Memory-Level Parallelism (MLP).
Domain Acceleration & Systolic Arrays	2D Systolic array dataflows (weight-stationary vs. output-stationary), PE grid scaling vs. SRAM bandwidth limits, RISC-V Vector (RVV) extensions.
Hardware Security & Side-Channels	Speculative execution side-channels (Spectre/Meltdown), way-partitioned secure caches, hardware enclave memory encryption overheads.
Ambitious Stretch Projects [AMBITIOUS]	OoO CPU + coherent AI accelerator co-design in gem5; hardware speculation barrier mechanisms in O3CPU; multi-die chiplet NoC interposer simulation.